

MANDEEP CHAUHAN

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EDUCATIONAL QUALIFICATIONS:

- B.Tech. Computer Science | 2022 - 2026
- Senior Secondary Education | CBSE | 2021
- Higher Secondary Education | CBSE | 2019

TECHNICAL SKILLS:

- **Languages:** Dart, Kotlin, Java, JavaScript, Python, C
- **Backend & APIs:** FastAPI, REST API, SQLAlchemy, JWT Auth, CORS, Async Python
- **ML/AI:** PyTorch, mBERT, scikit-learn, Librosa, Hugging Face, yt-dlp
- **Mobile:** Flutter, Android SDK, Jetpack Compose, Provider (State-management)
- **Databases:** PostgreSQL, Firebase (RTDB & Auth), Supabase
- **Tools and DevOps:** Git, GitHub, Android Studio, Gradle, Vercel, Render, Cloudinary, ImageKit

PROJECTS:

1. Multimodal music sentiment analysis and mood Transformation [Live Demo: vibe-lift.me]

PyTorch | mBERT | scikit-learn | Librosa | FastAPI | Next.js | Supabase | Cloudinary | Vercel

- Built a multilingual music sentiment system using text and audio features from Hindi songs.
- Created a human-annotated Bollywood valence dataset with songs for model training.
- Designed a multimodal architecture combining mBERT embeddings and Librosa audio features.
- Developed FastAPI inference APIs and deployed the full-stack platform using Vercel, Hugging Face, Cloudinary, and Supabase.

2. Social Media Web App [[GitHub: github.com/Mandeep347/social-media-app](https://github.com/Mandeep347/social-media-app)]

React.js | FastAPI | Tailwind CSS | SQLAlchemy | PostgreSQL | JWT Auth | ImageKit | Render | Vercel

- Developed a full-stack social media platform with authentication, media uploads, feeds, and user profiles.
- Built REST APIs with FastAPI and async SQLAlchemy integration.
- Implemented JWT authentication, password hashing, input validation, and optimized async operations.

3. YouTube Sync Playback App [[GitHub: github.com/Mandeep347/harmonique](https://github.com/Mandeep347/harmonique)]

Flutter | Dart | Firebase RTDB | YouTube Player

- Built a real-time synchronized YouTube playback app supporting multi-device room sessions.
- Implemented playback synchronization using Firebase timestamps and delay compensation.
- Enabled host-controlled play, pause, and seek operations across connected clients.

ACHIEVEMENTS:

- Led a 4-member final-year project team building a multimodal music sentiment analysis platform using FastAPI, PyTorch, and Next.js.
- Runner-up – Technical Quiz Competition at NITRA Technical Campus, 2025
- Runner-up – Coding Competition at NITRA Technical Campus, 2025